

AYUSHI RAJ

AI-Powered Full Stack Software Engineer | Backend & Systems Specialist

Bihar, India | ayushi29507@gmail.com | ayushiraj.me | github.com/Silenttears-cloud | linkedin.com/in/ayra404

PROFESSIONAL SUMMARY

Systems-focused Software Engineer with deep expertise in building low-latency REST APIs, multi-model LLM orchestration gateways, and full-stack web applications. Proficient across TypeScript, Node.js, Python FastAPI, C#/.NET Core, and Java Spring Boot, with a proven track record of architecting zero-knowledge cryptographic vaults and automated developer sandboxes. Demonstrated mastery in database normalization (3NF), Redis token-bucket caching, and automated testing (Jest, PyTest, xUnit) to deliver robust software systems.

TECHNICAL SKILLS

Backend & Systems: Node.js • Express.js • Fastify • Python (FastAPI) • C# (.NET Core) • Java (Spring Boot) • REST APIs

Databases & Caching: Redis • PostgreSQL • MongoDB • SQLite • Supabase • Schema Design (3NF)

Languages: C • Rust • TypeScript • JavaScript • Python • C# • Java • SQL • HTML5 • CSS3

Frontend UI & Full Stack: React.js • Next.js • Express.js • Redux • Monaco Editor • TailwindCSS • Bootstrap

Testing & DevOps: Jest • PyTest • xUnit • Moq • JUnit 5 • Docker • CI/CD • GCP • Git • Vercel • Render

AI Gateway & Security: Multi-Model Routing • Prometheus Telemetry • Web Crypto API (AES-GCM) • PBKDF2 • JWT Auth

TECHNICAL PROJECTS

Axiom AI Gateway - Low-Latency Multi-Model LLM Gateway (TypeScript, Node.js, Express, Redis, Docker)

[Live Demo](#) | [GitHub](#)

- **Product Architecture:** Architected a resilient LLM orchestration gateway that routes AI requests across OpenAI, Anthropic, and Gemini API endpoints to eliminate provider downtime.
- **Latency & Concurrency:** Designed an in-memory Redis token-bucket rate limiter and response cache, decreasing downstream provider API latency by 45% under high request volume.
- **Reliability & Telemetry:** Integrated Prometheus telemetry metrics and structured JSON audit logging to maintain 99.9% gateway uptime with zero silent request drops.

Alyra Lock - Secure Zero-Knowledge Password Vault (React, TypeScript, Express, MongoDB)

[Live Demo](#) | [GitHub](#)

- **Product Architecture:** Engineered a client-side zero-knowledge password vault ensuring master encryption keys remain entirely isolated inside user browser memory, mitigating cloud database leak risks.
- **Cryptographic Engineering:** Implemented Web Crypto API primitives utilizing AES-GCM (256-bit) payload encryption and PBKDF2 key derivation over 100,000 hashing iterations for key security.
- **Performance & Security:** Developed high-throughput Express.js REST API sync endpoints with JWT authentication and strict CORS headers, keeping vault database sync latency under 200ms.

Astra Vision - AI Sandbox & Code Parsing Platform (FastAPI, Python, Monaco, Tree-Sitter)

[Live Demo](#) | [GitHub](#)

- **Product Architecture:** Architected an automated code-parsing platform and browser sandbox that allows developers to safely execute untrusted code while visualizing full-stack repository dependency graphs.
- **Sandbox Runtime Challenge:** Built a self-healing Python execution engine with subprocess isolation and security checkpoints to intercept unauthorized OS file system calls and network socket requests in real time.
- **Indexing Performance:** Integrated Tree-Sitter AST compilers and ChromaDB vector search to parse repository syntax nodes, enabling instant code graph dependency analysis under 200ms.

TECHNICAL SIMULATIONS & VIRTUAL EXPERIENCES

Commonwealth Bank (CommBank) - Software Engineering Virtual Simulation

June 2026

- **Problem & Scope:** Diagnosed and resolved silent data overwrite bugs across high-traffic C#/.NET Core financial API controllers handling MongoDB document updates across financial services endpoints.
- **Action Taken:** Extended C# Web API controllers with `$set` atomic operators for partial payload updates and modernized an interactive React/Redux Goal Manager UI component for banking clients.
- **Engineering Outcome:** Authored comprehensive automated test suites using xUnit and Moq, covering complex boundary conditions to guarantee high reliability and transactional consistency.

Y Combinator (YC Startup - Shiptivity) - Software Engineering Virtual Simulation

June 2026

- **Problem & Scope:** Fixed key constraint collisions and slow re-rendering performance on a real-time drag-and-drop Kanban workflow board used by engineering teams to track task state transitions.
- **Action Taken:** Architected dynamic SQLite priority reordering logic using atomic transactions to update priority ranks sequentially and integrated Dragula drag-and-drop state hooks in React.
- **Engineering Outcome:** Reduced UI component re-render cycles by 30% and patched OpenSSL compilation bottlenecks in legacy Node.js Webpack configurations to optimize runtime builds.

Walmart USA - Advanced Software Engineering Virtual Simulation

May 2026

- **Problem & Scope:** Solved indexing overhead and performance bottlenecks in high-volume retail inventory processing queues handling unnormalized data ingestion from legacy spreadsheets.
- **Action Taken:** Implemented a generic K-ary Max Heap structure in Java using fast bitwise shift operators (`<<`, `>>>`) and built 3NF database schemas with design pattern abstractions.
- **Engineering Outcome:** Accelerated queue indexing calculations by 35% and engineered automated Python ETL pipelines using `csv` and `sqlite3` to clean and consolidate disparate spreadsheets.

EDUCATION

Bachelor of Computer Applications (BCA) - Amity University Noida

Expected July 2028

- **Relevant Coursework:** Data Structures & Algorithms (DSA) • Database Management Systems (DBMS) • Operating Systems (OS) • Computer Networks • Object-Oriented Programming (OOP)